Natural-language cell analysis on time-lapse microscopy: detailed architecture

Two trained models (Cellpose-SAM generalist, ViT-S classifier) + rule-based tracking & measurement + tool layer + LLM planner

A Perception: from pixels to a cell table (runs once per movie)

0 Input & data

CTC 2D+t: DIC-HeLa, SIM+,
N2DL-HeLa, U373
seq 01 = train / select
seq 02 = test
masks: gold, else silver
lineage: TRA + man_track

1 Generalist Cellpose-SAM

TRAINED

encoder: SAM ViT-L image encoder
head: 2 flow maps (dy,dx) + cell prob.
masks: follow flows to cell centres
fine-tuned from 'cpsam' on seq 01 of
ALL 4 datasets (321 train / 35 val)
100 ep, lr 1e-5, wd 0.1, batch 8
inference: cellprob 0, flow 0.4, tiles 256

2 Tracker (rules)

IoU between frames t-1, t
Hungarian matching, keep IoU >= 0.2
division: >=2 new cells within
1.5 parent diameters -> 2 daughters,
parent track ends at t-1
output: CTC masks + res_track.txt

3 Measurements

area, perimeter, eccentricity
solidity, circularity 4piA/P^2
irregularity, aspect ratio
mean intensity, texture CV
centroid (x right, y down)
track: begin, end, parent, kids

4 ViT-S classifier

TRAINED

vit_small_patch16_224 (ImageNet)
input: [image, image, cell mask]
crop 2.5 x cell diameter -> 224
classes: dividing / newborn /
interphase (labels from lineage)
15 ep, AdamW 1e-4, weighted CE
+ prior correction (real freq.)

CELL TABLE (one row per cell per frame)

id, frame, area, shape..., cx, cy, track begin/end/parent/daughters, p_dividing, p_newborn

reads

B Language: from a free-text question to a computed answer (runs per question)

User prompt

"Which 2 biggest cells
are in the lower half
of frame 12?"

free text, any wording

6 Planner: Qwen2.5-7B-Instruct

NOT TRAINED

open-weight LLM, run locally (vLLM)
OpenAI-compatible API on the job GPU
system prompt: image size, frames,
word->feature map (large=area,
round=circularity, lower half=y>H/2)
tool use, <= 15 steps, ends with submit

calls

results

5 Tool layer (state: frame + selection)

set_frame(t) select_all_cells() count()
filter(conditions | percentiles 'p75' |
region | box)
rank(feature, asc/desc, k) get_cell(id)
track(ids) divisions(start, end)
list_features() submit_answer(...)

Answer

set_frame(12) -> select_all_cells()
-> filter(region='lower')
-> rank('area','desc',2)
-> submit_answer([14, 17])

every number computed from the cell table

no API key, no internet, no cost: the LLM runs on the same GPU node as the pipeline

C Evaluation on sequence 02 (thesis results)

Oracle (reference masks)

reference annotation through the same
pipeline: dense silver (gold if none)
+ TRA lineage = upper bound
error attribution:
planner loss = 1 - oracle score
segmentation loss = oracle - generalist

Segmentation & tracking

SEG vs gold (CTC: Jaccard, >50% rule)
DET vs TRA markers, all frames
official TRA / DET / SEG (py-ctcmetrics)
per-frame SEG (diagnosis)
classifier: F1, AUROC, AUPRC

Language benchmark

~320 questions from seq-02 reference
8 types: top-k region, superlative, counts,
box, above median, divisions, about to divide
4 wordings: template, paraphrase,
LLM-written free text, hand-written
match cells by IoU >= 0.5 (Hungarian)

Reported

per dataset x backend (oracle / generalist):
selection F1, exact match
count relative error
division-event P / R
PNG figures + report.md
+ backup snapshot